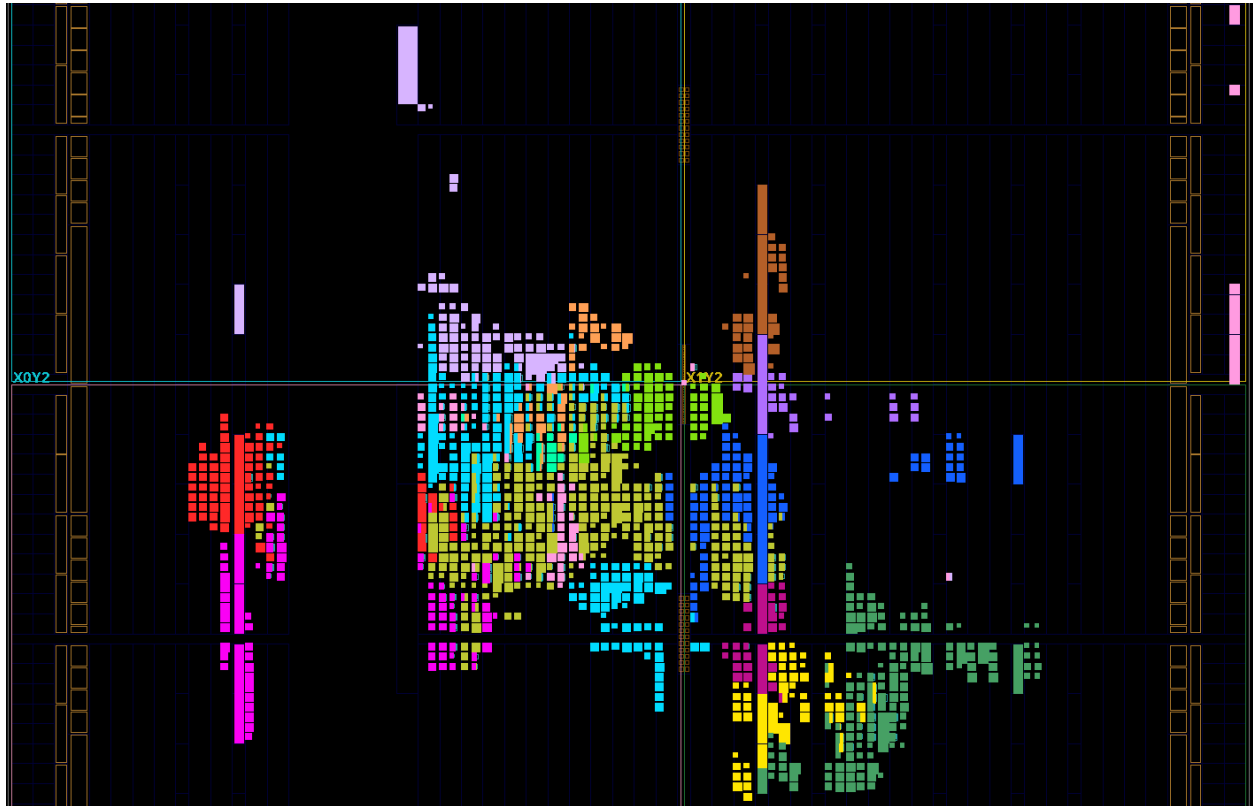


Target board: Artix A7-100T **Target FPGA:** XC7A100TCSG324-1 **Target clock:** 100 MHz

Worst negative slack: 0.491ns **Worst hold slack:** 0.010ns

Synthesis strategy: default, **Implementation strategy:** Performance_ExplorePostRoutePhysOpt



Name	Slice LUTs (63400)	Slice Registers (126800)	F7 Muxes (31700)	F8 Muxes (15850)	Slice (15850)	LUT as Logic (63400)	DSPs (240)	Bonded IOB (210)	BUFGCTRL (32)	XADC (1)
motor_controller_top	3362	2654	64	32	1210	3362	42	16	1	1
u_clarke (clarke)	83	56	0	0	28	83	6	0	0	0
u_clarke_inverse (clarke_inverse)	179	112	0	0	77	179	3	0	0	0
u_cordic (cordic)	225	100	0	0	71	225	0	0	0	0
u_current_cloop (current_cloop)	293	149	0	0	114	293	8	0	0	0
u_current_reader (motor_current_reader)	221	188	0	0	90	221	2	0	0	1
u_encoder (as5048a_spi_master)	167	66	0	0	64	167	0	0	0	0
u_park (park)	70	34	0	0	24	70	4	0	0	0
u_park_inverse (park_inverse)	74	38	0	0	25	74	4	0	0	0
u_parser (parser)	141	100	0	0	64	141	0	0	0	0
u_position_cloop (position_cloop)	248	149	0	0	93	248	4	0	0	0
u_pwm_driver (pwm_3ph_driver)	63	118	0	0	51	63	0	0	0	0
u_register_file (register_file)	704	553	64	32	343	704	0	0	0	0
u_serializer (serializer)	25	64	0	0	18	25	0	0	0	0
u_svpwm (svpwm)	254	203	0	0	123	254	3	0	0	0
u_tracker (tracker)	242	306	0	0	131	242	0	0	0	0
u_uart_rx (uart_rx)	71	73	0	0	32	71	0	0	0	0
u_uart_tx (uart_tx)	28	60	0	0	18	28	0	0	0	0
u_velocity_cloop (velocity_cloop)	201	165	0	0	88	201	8	0	0	0

Setup Time Critical Paths

Intra-Clock Paths - clk - Setup

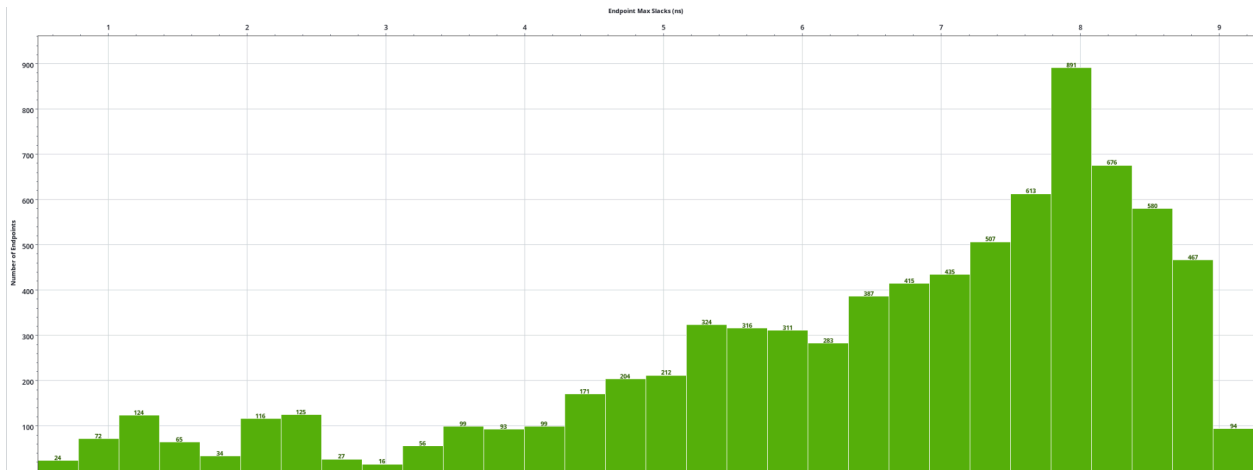
Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement
Path 1	0.491	6	16	u_park/ARG/CLK	u_park/o_d_reg[10]/D	9.423	6.987	2.436	10.000
Path 2	0.504	6	16	u_park/ARG/CLK	u_park/o_d_reg[11]/D	9.415	6.987	2.428	10.000
Path 3	0.508	6	16	u_park/ARG/CLK	u_park/o_d_reg[13]/D	9.407	6.987	2.420	10.000
Path 4	0.512	6	16	u_park/ARG/CLK	u_park/o_d_reg[11]/D	9.406	6.987	2.419	10.000
Path 5	0.593	6	16	u_park/ARG/CLK	u_park/o_d_reg[9]/D	9.323	6.998	2.325	10.000
Path 6	0.597	6	16	u_park/ARG/CLK	u_park/o_d_reg[8]/D	9.319	6.998	2.321	10.000
Path 7	0.604	6	16	u_park/ARG/CLK	u_park/o_d_reg[6]/D	9.310	6.998	2.312	10.000
Path 8	0.621	14	5	u_position_cloop...get_pos_reg[3]/C	u_position_cloop/ARG_2/RSTB	8.852	3.904	4.948	10.000
Path 9	0.632	6	16	u_park/ARG/CLK	u_park/o_d_reg[7]/D	9.287	6.998	2.289	10.000
Path 10	0.640	6	16	u_park/ARG/CLK	u_park/o_d_reg[0]/D	9.277	6.987	2.290	10.000
Path 11	0.644	6	16	u_park/ARG/CLK	u_park/o_d_reg[3]/D	9.273	6.987	2.286	10.000
Path 12	0.645	6	16	u_park/ARG/CLK	u_park/o_d_reg[2]/D	9.270	6.987	2.283	10.000

Hold Time Critical Paths

Intra-Clock Paths - clk - Hold

Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay
Path 1001	0.010	1	1	u_serializer/r_frame_reg[3][1]/C	u_serializer/o_byte_reg[1]/D	0.364	0.186	0.178
Path 1002	0.023	1	2	u_cordic/o_vid_reg/C	r_trig_fresh_reg/D	0.378	0.209	0.169
Path 1003	0.053	2	7	u_cordic/r_y_reg[13]/C	u_cordic/r_y_reg[16]/D	0.423	0.342	0.081
Path 1004	0.058	0	1	u_serializer/o_byte_reg[7]/C	u_uart_tx/r_byte_reg[7]/D	0.402	0.164	0.238
Path 1005	0.064	2	7	u_cordic/r_y_reg[13]/C	u_cordic/r_y_reg[18]/D	0.434	0.353	0.081
Path 1006	0.067	1	11	u_serializer/r_byte_cnt_reg[2]/C	u_serializer/o_byte_reg[4]/D	0.425	0.209	0.216
Path 1007	0.086	1	5	u_cordic/FSM_one...r_state_reg[0]/C	u_cordic/FSM_one...r_state_reg[1]/D	0.454	0.186	0.268
Path 1008	0.089	2	7	u_cordic/r_y_reg[13]/C	u_cordic/r_y_reg[17]/D	0.459	0.378	0.081
Path 1009	0.089	2	7	u_cordic/r_y_reg[13]/C	u_cordic/r_y_reg[19]/D	0.459	0.378	0.081
Path 1010	0.090	2	60	u_cordic/r_iter_reg[0]/C	u_cordic/r_y_reg[13]/D	0.473	0.251	0.222
Path 1011	0.092	3	7	u_cordic/r_y_reg[13]/C	u_cordic/r_y_reg[20]/D	0.462	0.381	0.081
Path 1012	0.094	2	60	u_cordic/r_iter_reg[0]/C	u_cordic/r_y_reg[12]/D	0.477	0.256	0.221

Slack Time Histogram



Power Usage Report

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power:	0.123 W
Design Power Budget:	Not Specified
Process:	typical
Power Budget Margin:	N/A
Junction Temperature:	25.6°C
Thermal Margin	59.4°C (12.9 W)
Ambient Temperature	25.0 °C
Effective θ_{JA}	4.6°C/W
Power supplied to off-chip devices	0 W
Confidence level	Low
Launch Power Constraint Advisor to find and fix invalid switching activity	

